

1. Project Title

EduMind AI — Intelligent Academic Advisor for University Students

2. Problem Statement

University students in Bangladesh, especially in large departments face several recurring academic challenges that remain unsolved by existing tools:

- Lack of personalized study planning: Students cannot effectively organize their study schedule according to their course load, deadlines, and personal learning pace.
- No instant academic guidance: Students often have to wait days to get answers from teachers or seniors, slowing their progress significantly.
- Poor resource discovery: Finding the right study materials, videos, or practice problems for a specific topic is time-consuming and unstructured.
- No performance tracking: Students have no easy way to monitor their own understanding and identify weak areas before exams.
- Manual quiz preparation: Creating practice quizzes from notes or textbooks is tedious and rarely done by students independently.

Existing generic tools like Google or YouTube lack academic context and personalization. No existing platform combines personalized study planning, AI-powered academic assistance, automated quiz generation, and performance analytics in a single, university-focused tool built for Bangladeshi students.

EduMind AI is designed to close all of these gaps in one integrated platform

3. Objectives

The primary objectives of this project are:

1. Build a secure, full-stack web application with JWT-based authentication and role-based access control.
2. Develop a real-time AI chatbot with full conversation memory that answers academic questions with contextual understanding.
3. Implement an AI quiz generator that creates practice questions from student-uploaded PDF or text notes.
4. Create an AI study planner that generates personalized weekly schedules based on course load, available time, and deadlines.
5. Build a smart resource recommendation engine that suggests relevant videos, articles, and practice problems.
6. Implement an AI-powered performance analytics dashboard with visual charts and automatic weak-area detection.
7. Add bilingual AI support (English + Bangla) so students can interact in their preferred language.
8. Build a Pomodoro-integrated focus timer with AI-suggested study session lengths based on topic complexity.
9. Deploy the complete application to a live, production-ready URL with proper environment variable management.
10. Maintain full technical documentation, a well-organized GitHub repository, and weekly progress commits throughout development.

4. Feature Set

4.1 Basic Feature Set

#	Feature	Description	AI?
1	User Authentication	Secure register, login, logout with JWT + bcrypt password hashing	No
2	Student Dashboard	Central hub showing study streaks, upcoming tasks, AI tips, and quick access to all features	No
3	Task & Deadline Tracker	Add courses, assignments, and exam dates with visual countdown timers	No
4	Note Management	Upload, organize, and retrieve notes by subject; supports PDF and text files	No
5	Admin Panel	Admin can view all users, activity logs, and platform-wide statistics	No
6	Responsive UI	Fully responsive design working on desktop, tablet, and mobile	No
7	Profile & Settings	Student profile with avatar, course list, notification preferences	No

4.2 AI-Powered Features

#	AI Feature	How It Works	Impact
1	Contextual AI Chatbot	Full conversation history sent to GPT-4o-mini. Remembers previous messages within a session and across sessions using stored context.	Students get coherent, context-aware academic help
2	AI Quiz Generator	Student uploads PDF notes. Backend extracts text, sends to AI with structured JSON prompt. Returns MCQs + short answers rendered as interactive quiz.	Eliminates manual quiz creation; exam-readiness improves
3	AI Study Planner	Student inputs subjects, daily hours, exam dates. AI returns a structured weekly timetable prioritizing high-risk subjects.	Personalized planning without manual scheduling effort
4	Resource Recommender	AI analyzes the student's current topic and chat history, then recommends specific YouTube channels, articles, and practice sites with brief explanations.	Saves hours of resource hunting
5	Weak Area Detector	AI analyzes quiz scores and chatbot question history to identify which topics the student struggles with most and surfaces them on the dashboard.	Proactive guidance before exams
6	Bilingual Support	Students can switch the chatbot language to Bangla. System prompt instructs AI to respond in Bangla with academic vocabulary.	Removes language barrier for local students
7	AI Flashcard Generator	AI converts uploaded notes into spaced-repetition flashcards. Students flip through cards with a built-in self-rating system.	Proven memorization technique, automated
8	Pomodoro + AI Timer	AI recommends optimal Pomodoro session lengths based on topic complexity (e.g., 25 min for revision, 50 min for new complex topics). Tracks focus time.	Improved concentration and study efficiency

5. Technology Stack

The following technology stack has been selected based on industry relevance, learning resources, and feasibility within the semester timeline:

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	User interface, dashboard, forms, and responsive design
Backend	Node.js + Express.js	REST API development, business logic, authentication
Database	PostgreSQL	Storing users, notes, tasks, quiz results, and activity logs
Authentication	JWT + bcrypt	Secure token-based login and password hashing
AI Service	OpenAI API (GPT-4o-mini) / Gemini	Chatbot responses, quiz generation, study plan suggestions
File Handling	Multer + PDF-parse	Handling note uploads and extracting text from PDFs
Frontend Deploy	Vercel	Free and fast deployment of React frontend
Backend Deploy	Render	Free backend hosting for Node.js/Express server
Database Host	Supabase (PostgreSQL)	Managed cloud database with free tier
Version Control	GitHub	Source code management, collaboration, commit tracking

6. Expected Outcome

Upon successful completion, EduMind AI will deliver:

A live, publicly accessible full-stack web application deployable from any browser

Eight integrated AI features demonstrating genuine, meaningful use of large language models

A streaming AI chatbot with persistent memory the most technically impressive feature for viva

An automated quiz and flashcard system with spaced repetition logic

A visual analytics dashboard with auto-populated charts requiring no student manual input

Bilingual support (English + Bangla) a unique, locally relevant innovation

A well-documented, consistently committed GitHub repository demonstrating industry-standard workflow

Complete technical report, user manual, and API documentation